



SIMATS Online Programming Lab

DAA PROGRAMMING



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ASSESSMENT-CO3/AT1-EXPERIMENTS

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Language:

Python

C

C++

Java

QUESTIONS

Q1

Matrix Multiplication

10 marks



1/1 answered

Q1 Matrix Multiplication

10 marks

Evaluate divide and conquer matrix multiplication by implementing recursive splitting of matrices. Record execution time for different matrix sizes. Analyze how problem division reduces complexity. Interpret results and justify the efficiency of recursive approaches.

Your Code

```
1 def add(A, B):
2     n = len(A)
3     return [[A[i][j] + B[i][j] for j in range(n)] for i in range(n)]
4
5 def multiply(A, B):
6     n = len(A)
7
8     if n == 1:
9         return [[A[0][0] * B[0][0]]]
10
11     m = n // 2
12
13     A11 = [row[:m] for row in A[:m]]
14     A12 = [row[m:] for row in A[:m]]
15     A21 = [row[:m] for row in A[m:]]
16     A22 = [row[m:] for row in A[m:]]
17
18     B11 = [row[:m] for row in B[:m]]
```

```
19     B12 = [row[m:] for row in B[:m]]
20     B21 = [row[:m] for row in B[m:]]
21     B22 = [row[m:] for row in B[m:]]
22
23     C11 = add(multiply(A11, B11), multiply
24     C12 = add(multiply(A11, B12), multiply
25     C21 = add(multiply(A21, B11), multiply
26     C22 = add(multiply(A21, B12), multiply
27
28     return [C11[i] + C12[i] for i in range
29
30 A = [[1, 2],
31       [3, 4]]
32
33 B = [[5, 6],
34       [7, 8]]
35
36 C = multiply(A, B)
37
38 for row in C:
39     print(row)
```



Input

stdin input



Output

```
[19, 22]
[43, 50]
```

✓ Submitted